

Cells as the basic unit of life





Overview

Cell Theory

Cell Structure

Cell membrane

Cytoplasm (organelles)

Nucleus

Differences between plant and animal cells



Cell Theory

**All living organisms are
made of cells**

**Cells are the basic unit
of living things**

**All cells come from
other cells**



Cells, Tissues, Organs and Systems

The Organisation of Living Things

Click on each arrow in turn to see more:

Cells

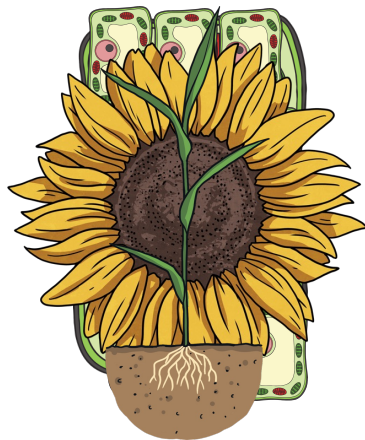
Tissues

Organs

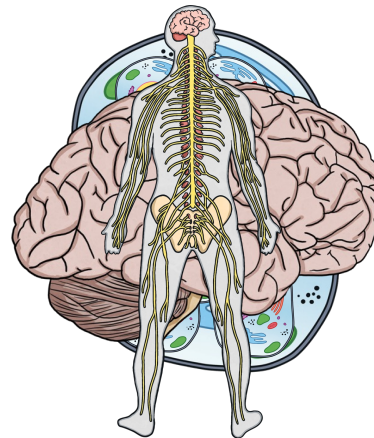
Organ Systems

Organisms

Plant



Animal



Systems are a group of organs working together.

Cell Theory



There are 2 types of organisms based on their cell structure:

Eukaryotic

These cells have a nucleus; they're found in animals, plants, fungi and protists. **YOU** are **Eukaryotic**!

Prokaryotic

These cells have no nucleus; they're found in bacteria and archaea

The human body is made of different cells that perform different functions.

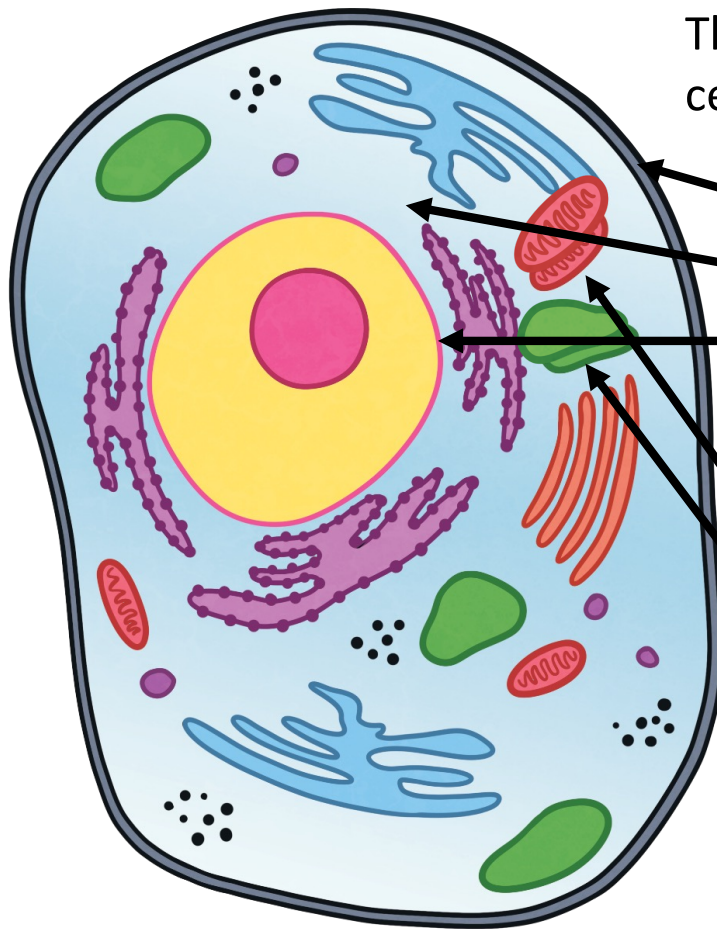
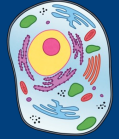
Cells vary in shape and size depending on their function.

Cells are microscopic and can only be seen under a microscope. Each cell is made of many many atoms arranged in complex molecules.



Cell Theory

All cells come from existing cells.



Though types of cells vary, some components of cell structure are common in all cells:

- cell membrane
- cytoplasm
- nucleus (in eukaryotic cells)

Some organelles are also common in cells:

- Mitochondria
- vacuoles



Cell Structure

Cytoplasm

Cytoplasm consists of cytosol and cell organelles:

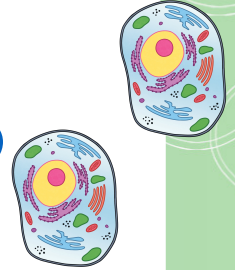
Cytosol is a colourless, watery, jelly substance (70–90% water)

A mixture of substances dissolved in water, e.g. salts, potassium, sodium, and proteins

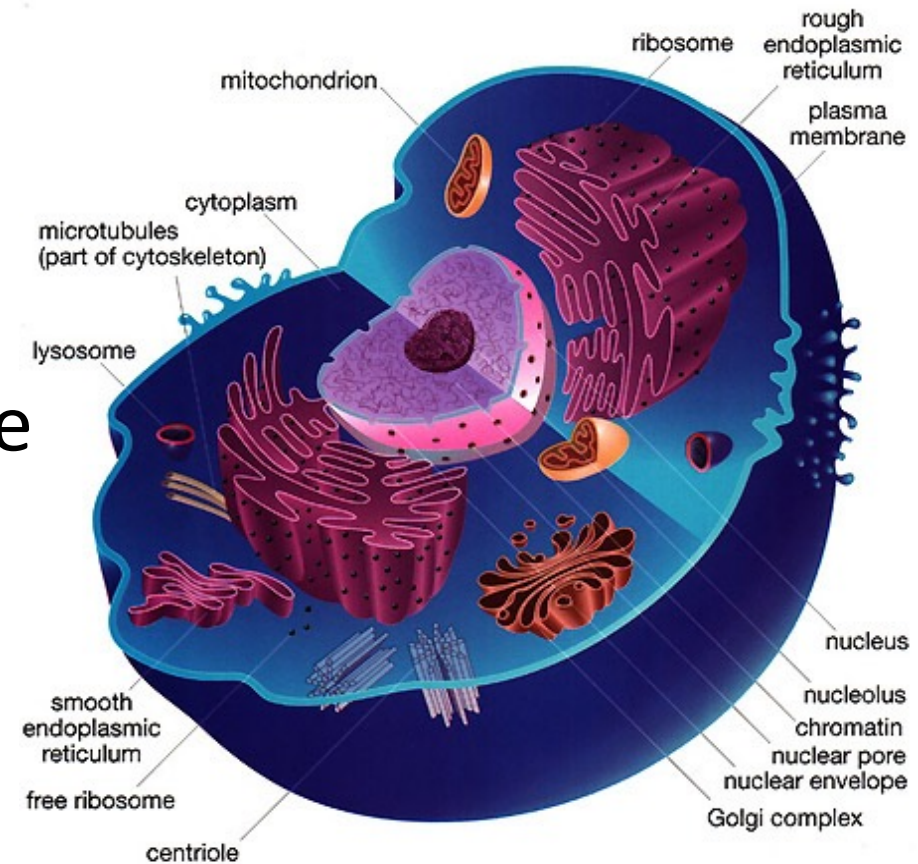
Cytosol is also where chemical reactions take place.

- **Glue** of the cell

Cell Organelles



- Organelle= “little **organ**”
- Found only inside **eukaryotic cells**
- Each organelle has a special job that helps the cell function
- Cells with different functions may have different organelles



BIOLOGY: Life on Earth, Fifth Edition
by Teresa Audesirk and Gerald Audesirk

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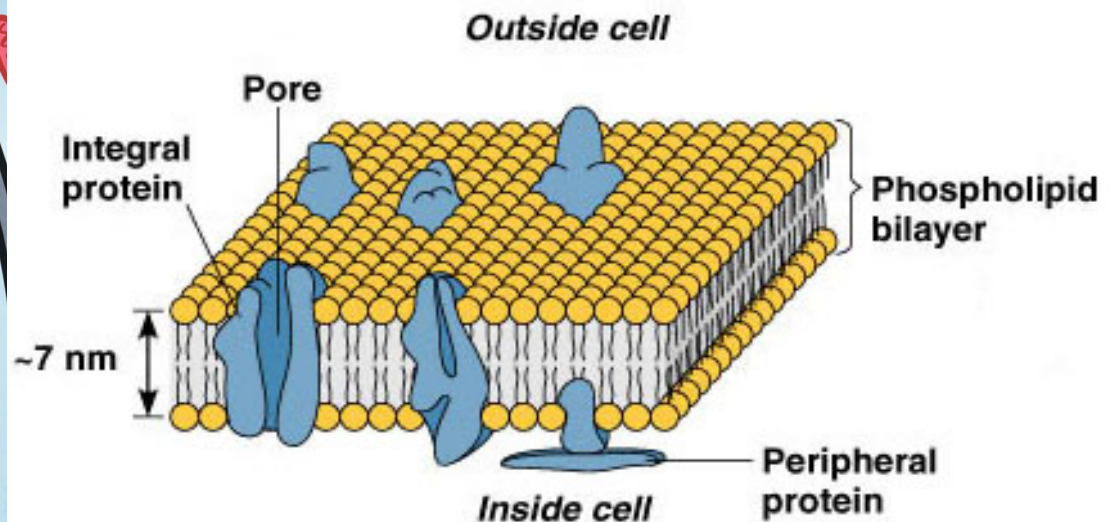
Cell Structure

Cell Membrane

Cell Membrane

a thin layer around the cell that separates it from its environment

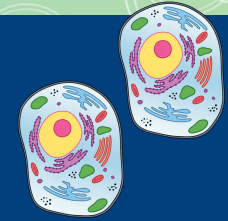
- **Gate** of the cell
- Made of a phospholipid bilayer
- Using chemistry, it controls what can enter the cell



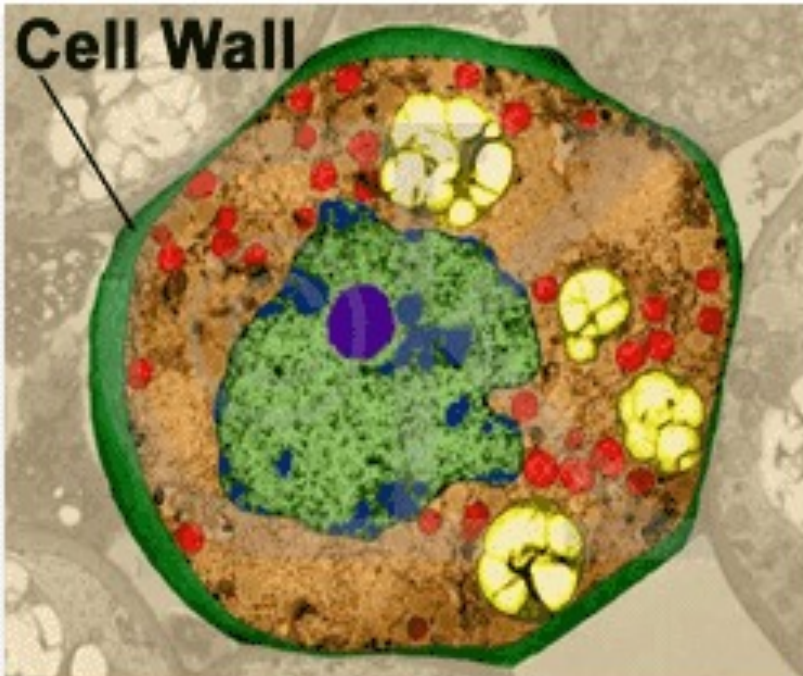
(b) Phospholipid bilayer of membrane



Cell Wall



Cell Wall



Found in plant cells,
never in animal cells

Extra layer on the outside of the cell
membrane

Gives plant cell support and stability

Cell wall contains cellulose: helps to keep the
shape of the plant cell

- **City Wall** of the cell

The shape of the plant cell helps to keep
the plant upright and rigid.

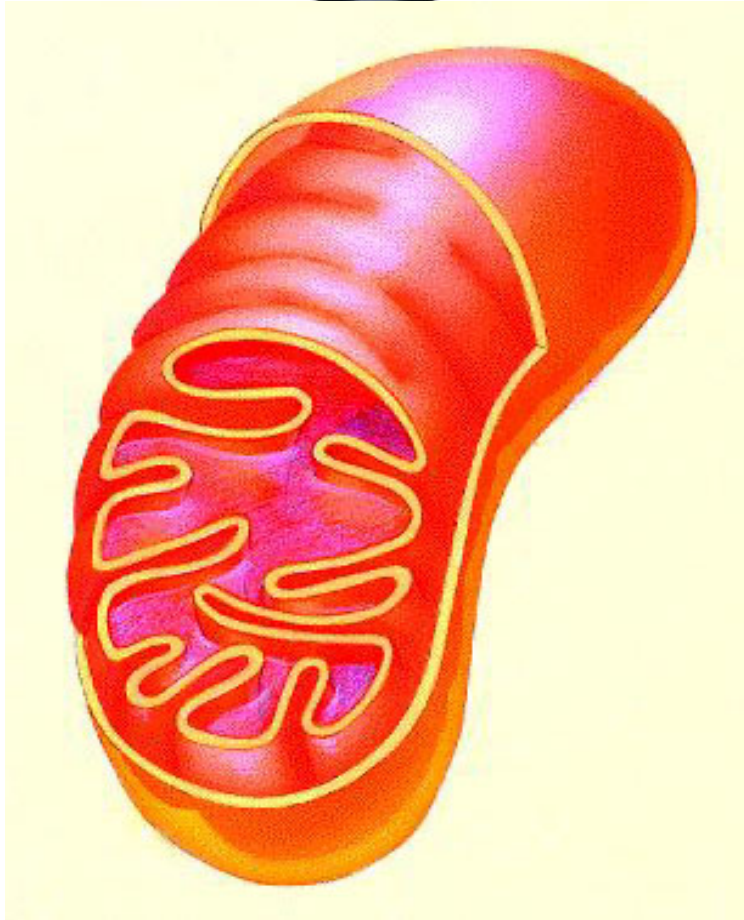


Mitochondria

Organelles

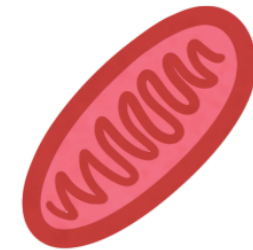


- “Power Plant of the cell”



Surrounded by a double membrane
Has its own strand of DNA

Active cells have more
mitochondria than
inactive cells

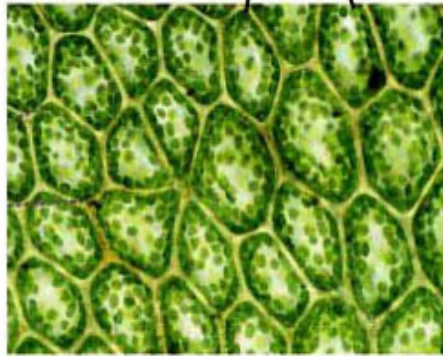
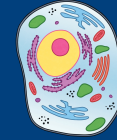


Cellular respiration occurs here to release
energy (ATP) for the cell to use

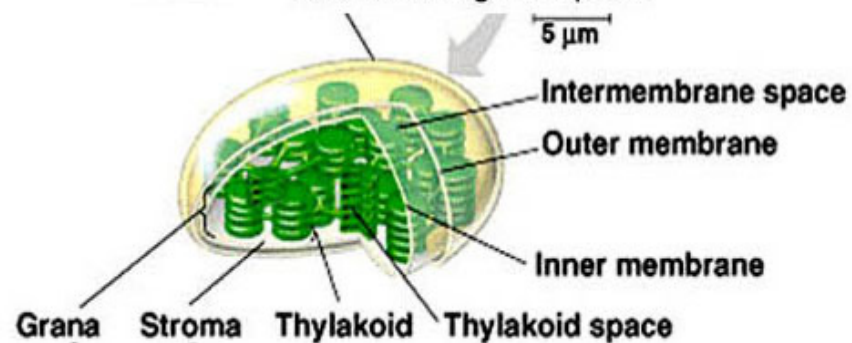
Carbon dioxide, water, and waste = the result
of this process

Chloroplasts

Organelles



Cells containing chloroplasts

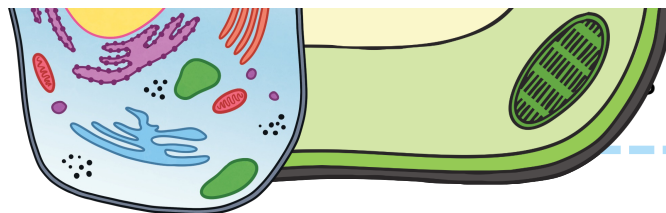


Only present in cells of plants, algae and a few species of amoeba

The only organelle that can produce food from the sun's energy

Contains **chlorophyll** – enables photosynthesis to take place

- Cell's **Solar Panel**



don't have chloroplasts and can't photosynthesise



Vacuoles

Organelles



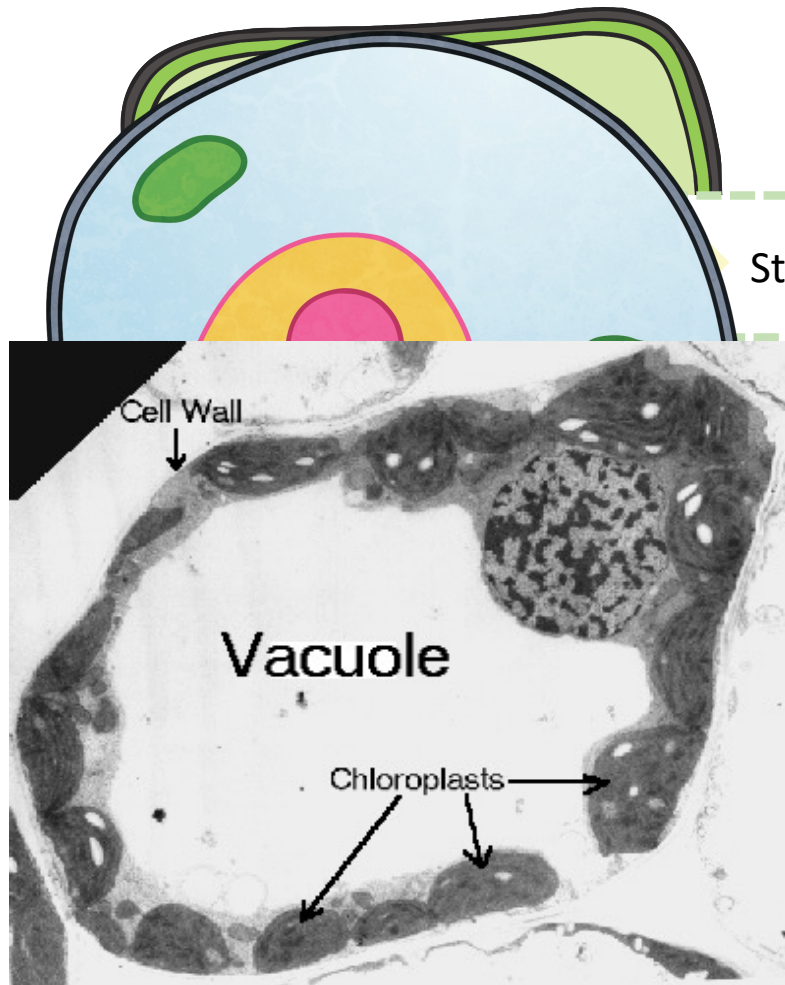
Important in plant cells

Stores liquid, food, waste and other materials

The cell's **warehouse**

Full Vacuoles help the plant stay rigid: leaves and stem are firm, upright

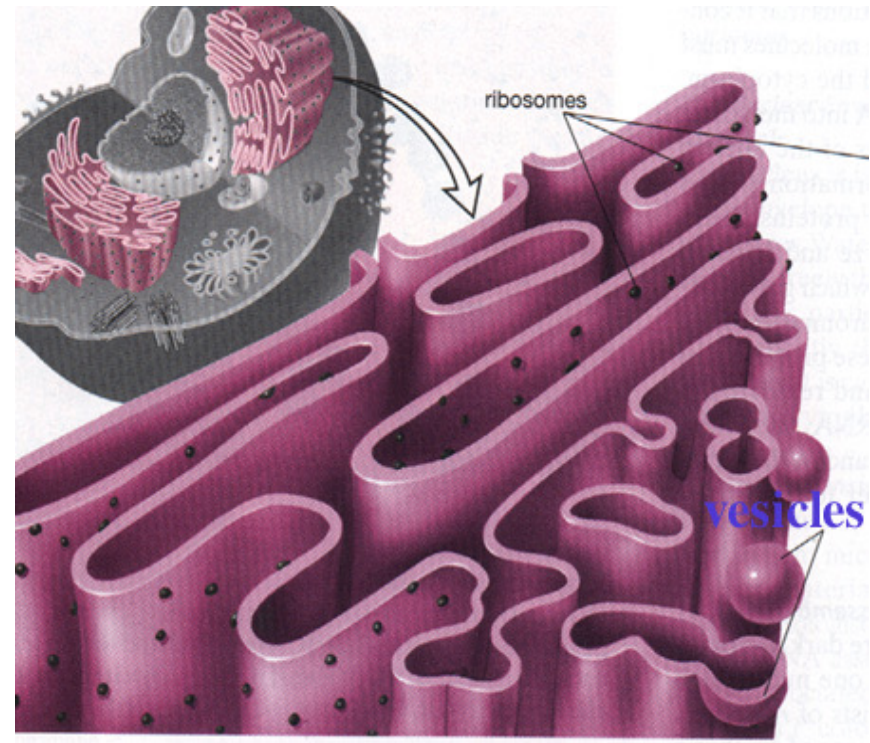
Vacuoles are present in some animal cells but are smaller with a shorter lifespan.



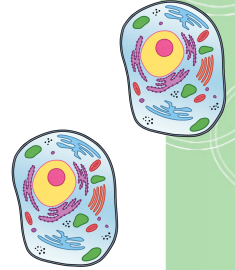
Endoplasmic Reticulum (ER)



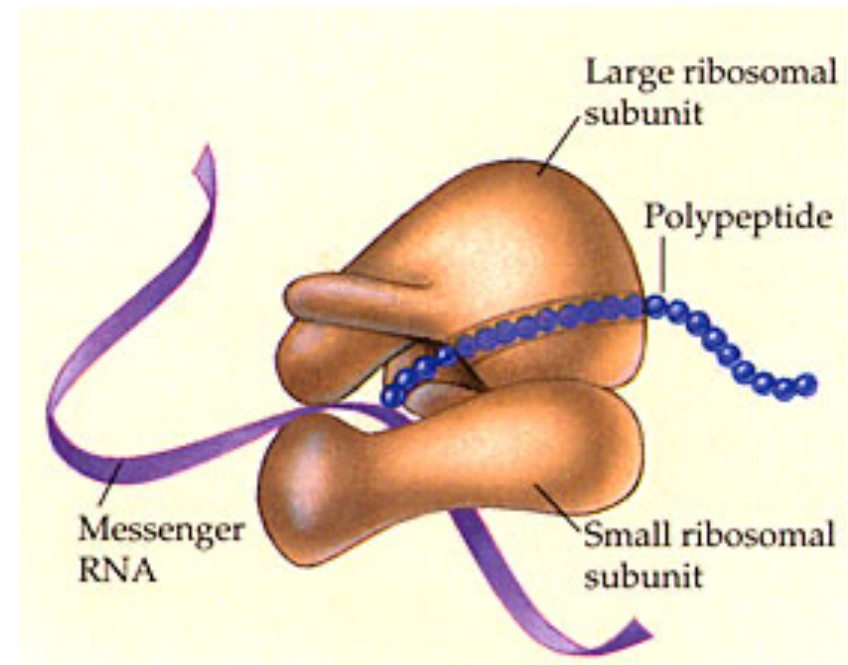
- A.k.a. “ER”
- Transports materials
- **Railroad** of the cell
- **Rough ER**: has ribosomes on it; it makes proteins
- **Smooth ER**: no ribosomes; it makes lipids



Ribosome



- **Construction Site** of the cell
- Builds proteins
- Found attached to rough ER or floating free in cytosol
- Produced in a part of the nucleus called the **nucleolus**



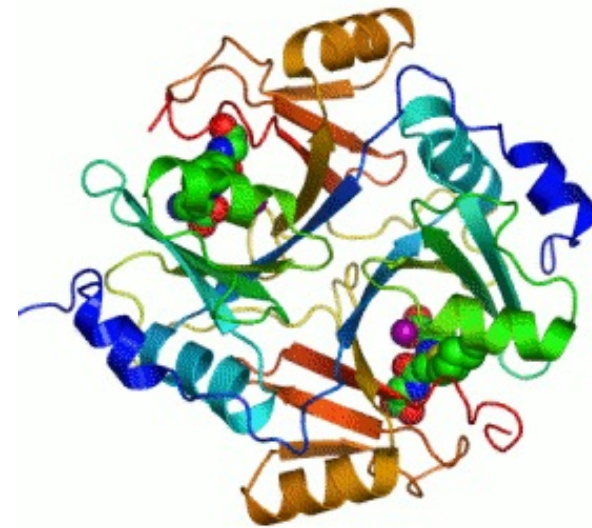


Proteins

Special molecule machines



- Large molecules with a specific shape and chemistry
- Different proteins help the cell do almost everything
- Proteins are built with instructions from DNA
- The cell's **machines**
- Made from amino acids

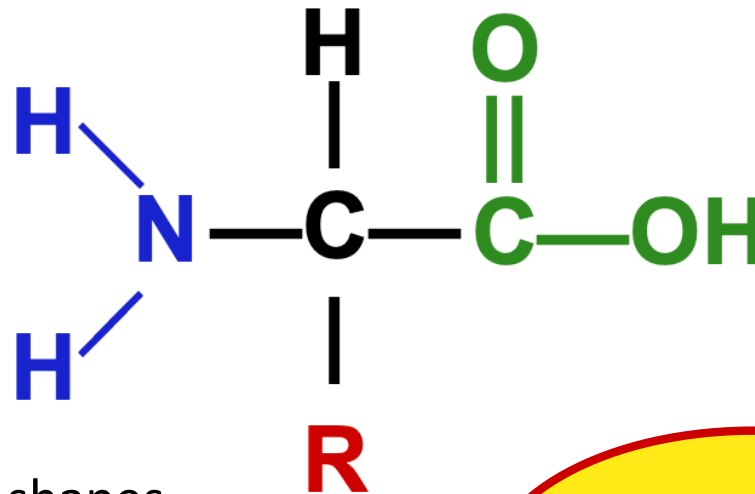


Amino Acids

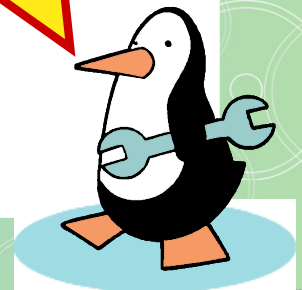
Building blocks of proteins

- Structure

- central carbon
- amino group
- carboxyl group (acid)
- R group (side chain)
 - can take many different shapes
 - helps each type of amino acid do its job
 - like 20 different letters of an alphabet
 - can make many words (proteins)



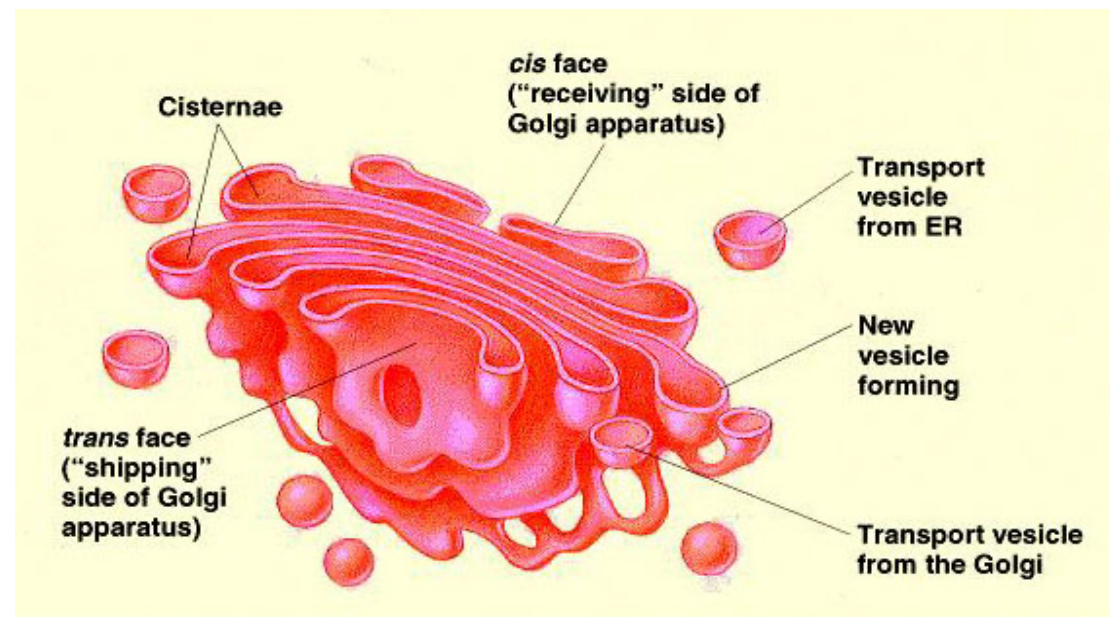
Oh, I get it!
amino = NH_2
acid = COOH



Golgi Apparatus



- **Post Office** of the cell
- Looks like a stack of plates
- Stores, modifies and packages proteins



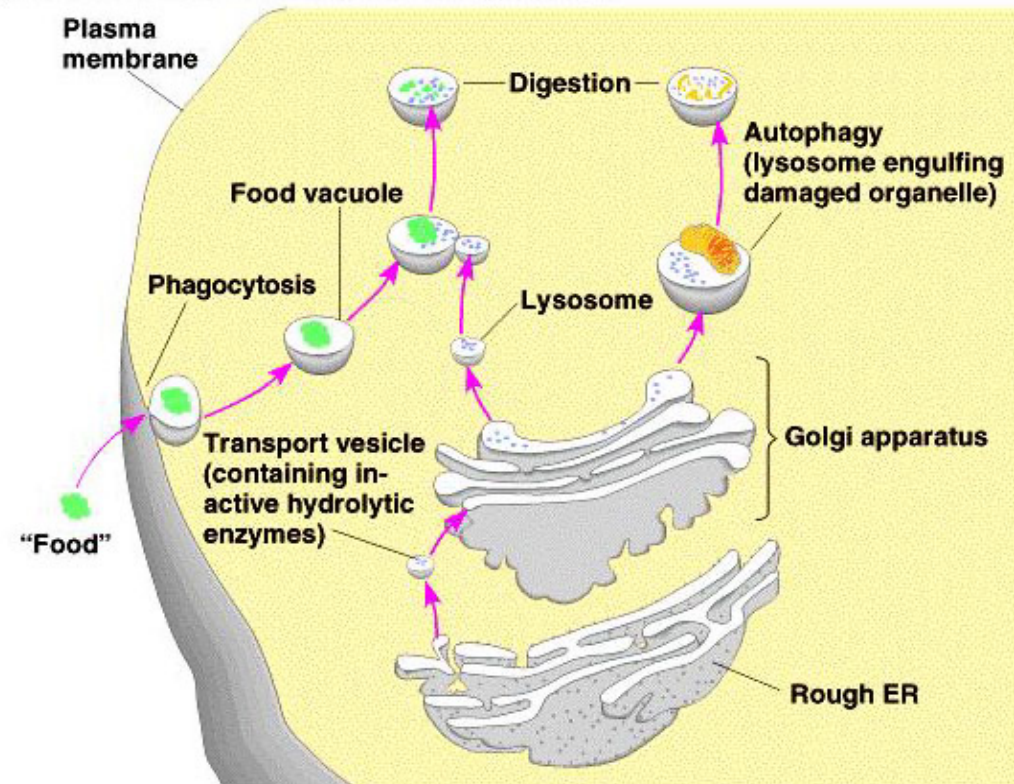
Lysosomes



- **Recycling Center** of the cell
- Contains digestive enzymes that break down waste



Figure 7.14 Formation and functions of lysosomes



Which organelles do lysosomes work with?



Nucleus

Control Center of the cell

Plant and animal cells have a nucleus in the cytoplasm

The nucleus controls chemical reactions and processes in the cell.

Nucleus also contains the cell's genetic material (DNA molecules).

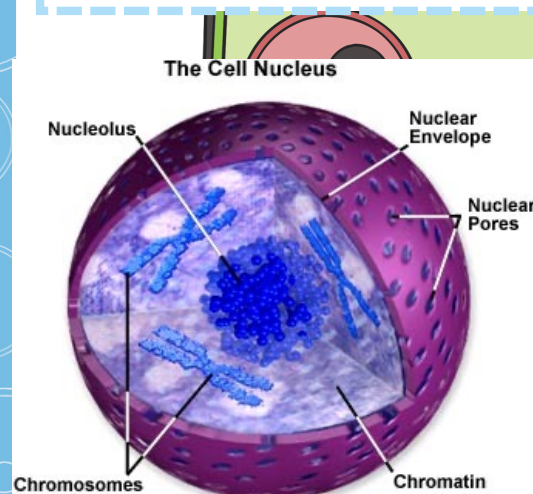


Figure 1

The structure of the nucleus:

Nuclear membrane = double membrane around the DNA

Nucleolus is inside the nucleus



Nucleus

DNA

DNA contains the instructions for how to build an organism

DNA contains info about hereditary characteristics.

- E.g. eye and skin colour, allergies, or illnesses passed down through generations

Differences in DNA = variation

DNA variation can cause illness in some organisms but can also help a species adapt to the environment.

Each Organism has unique DNA.

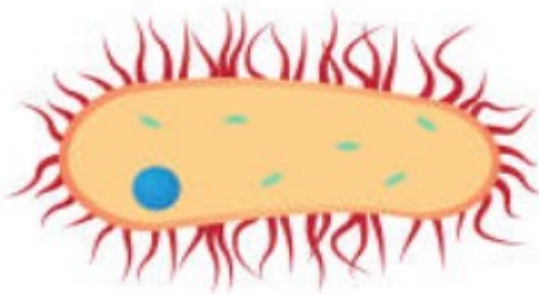


Cilia and Flagella



- Helps single-celled organisms move
- The cell's **leg** or tail

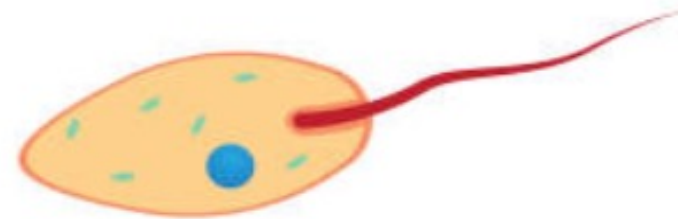
Cilia



Small, hair-like structure
responsible for locomotion

found only in eukaryotic cells

Flagella



Long, lashlike appendage serving
as an organ of locomotion

found in both prokaryotic and
eukaryotic cells

The Difference Between Plant and Animal Cells



Plant Cells

Contain plastids: chloroplasts, chromoplasts & leucoplasts

Have cellulose cell wall

Large central vacuole filled with cell sap

Small, irregular rigid shape

Nucleus is located at the edge of the cell

Animal Cells

Do not contain plastids

No cell wall

Few vacuoles (if any)

Large regular shape

Nucleus is located at the centre of the cytoplasm

